

WASP-50b

R-1598

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_141126 · created 2026-08-02T01:59:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456987.783 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.018
Transit depth (Rp/Rs)^2	1.71 +/- 0.46 [%]
Orbital Inclination (inc)	84.15 +/- 0.94
Airmass coefficient 1 (a1)	0.9658 +/- 0.0075
Airmass coefficient 2 (a2)	0.0213 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	#2 - [101, 421]
Optimal Aperture	3.93
Optimal Annulus	16.19
Transit Duration (day)	0.0684 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.018 vs 0.1404 (deviation 0.45σ)
Duration fit vs published	0.0684 d vs 0.0754 d (deviation 0.64σ)
Mid-time O–C	-3.7 min
Depth SNR	6.3
Residual scatter	0.78 %

Light curve

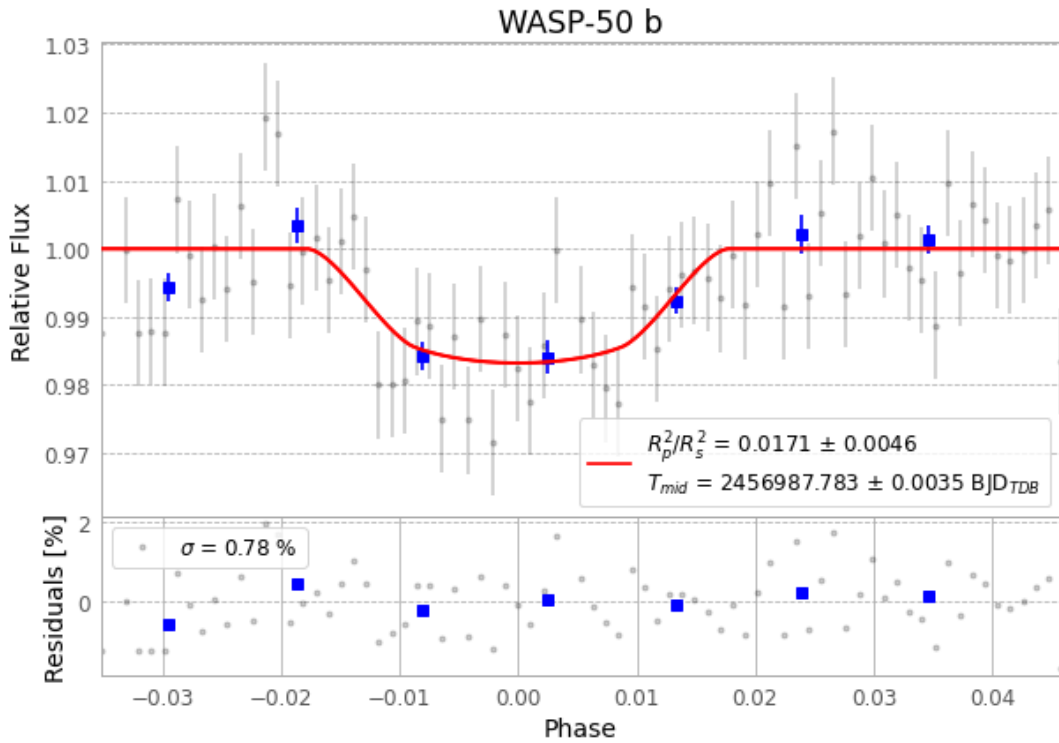


Figure 1 — FinalLightCurve_WASP-50 b_2014-11-25.png (SHA-256 1787404b58cec12c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d809ba618d188c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50141126050012.FITS → WASP-50141126084812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-141126.log (83be516207e0...), FinalLightCurve_WASP-50 b_2014-11-25.png (1787404b58ce...), FinalLightCurve_WASP-50 b_2014-11-25.csv (33accb465d6c...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:59Z.